The Finite Horizon: Variance Dominance, the Description–Inference Distinction, and Why Published Odds Describe the Process, Not the Participant

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Abstract

This paper formalizes and tests a simple thesis: *published odds are properties of a stochastic process (equivalently, of an infinite ensemble), and no finite-horizon participant ever lives at that horizon.* For i.i.d. trials with per-trial mean μ and standard deviation σ, a finite sample is variance-dominated below the *variance horizon* $n^* = (\sigma/\mu)^2$, where accumulated edge first outgrows accumulated noise. Below n^*, the probability of finishing on the wrong side of the edge approaches a universal large-n curve $\Phi(-\sqrt{n/n^*})$, verified by simulation across *seven* unrelated processes (biased coin, dice game, roulette, Gaussian walk, high-variance slot, -110 sports bettor, full blackjack basic-strategy engine) at up to 500,000 replications per point (Modelled, B), which collapse onto one line; the curve is asymptotic, with documented small-n lattice deviations checked against exact binomial probabilities. Two new robustness tests extend the claim beyond i.i.d. trials: Student-t increments ($\nu = 3, 5$) deviate from the curve by at most 1.4 percentage points (Modelled, B), and AR(1) trials collapse onto the same curve once n^* is computed from long-run variance. In the zero-edge limit the horizon is infinite: a fair coin's lead grows as $\sqrt{n}$ forever and one side leads most of the time (arcsine law), so "it evens out" holds only in relative, never absolute, terms. Optional stopping is shown to reshape but never remove expectation ($E[S_\tau] = \mu E[\tau]$ under standard regularity conditions). A 20,000-player cohort shows the volume asymmetry in full: the house measures its edge to ± 0.03pp at 10^7 bets while players at 200 bets see 26% winners, and the "advertised winner" (+50 units) is the predicted order statistic of the cohort ($\approx$+48), not evidence of skill. Finally, the applicability boundary is formalized: statistical noise is not a property of data but of the *question* asked of it: describing a closed finite universe (5/10 on a test) is exact, while inferring a latent process from a finite sample carries uncertainty shrinking only as $1/\sqrt{n}$ (quantified from $n = 10$ to 10^6). Whoever holds the volume lives at the process horizon and collects the edge; everyone else lives in the noise and pays the tax.

Keywords: finite-horizon statistics, variance dominance, signal-to-noise ratio, arcsine law, optional stopping, order statistics, law of small numbers

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1 Introduction

The published odds of a game (a 50/50 coin flip, blackjack basic strategy's ≈43% win / 9% push / 48% loss, roulette's −2.7% edge) are stated as properties of the game. The puzzle that motivated this work is that almost nobody experiences them. Flip a coin 100 times and the result is routinely skewed to one side. Sit at a blackjack table and the bankroll oscillates: up, then down, then up again. The textbook percentages are properties of the *process* (what happens over an effectively infinite horizon, or equivalently across an infinite ensemble of trajectories), but every actual participant, every session, every season, every career, is one finite trajectory, and a single finite trajectory need not resemble, even approximately, the process parameters it is drawn from.

The research question is therefore: *where, quantitatively, does the long run begin?* How many trials must elapse before the published odds become a useful description of any individual's experience, and what does the pre-convergence regime look like? A secondary question concerns scope: the finite-horizon argument cannot apply to everything (a 5/10 test score *is* 50%, exactly, with no noise involved), so where is its boundary?

The answers, developed theoretically and verified by simulation, are threefold. First, the crossover is governed by a single number, the variance horizon $n^* = (\sigma/\mu)^2$, and the pre-convergence regime is described by a universal *asymptotic* curve. Second, the zero-edge case is degenerate in an illuminating way: a fair coin never converges in absolute terms. Third, the noise that finite samples carry is not a property of the data but of the inferential question asked of it: a distinction with sharp, testable consequences.

The paper proceeds as follows. Section 2 develops the theory: the variance horizon, the universal wrong-side curve, the zero-edge limit, optional stopping, and the description–inference boundary. Section 3 describes the simulation battery, the evidence-grading vocabulary applied to every reported number, and the retrospective nature of the analysis. Section 4 reports the results test by test, including new robustness tests under fat-tailed and autocorrelated increments. Section 5 states what the results mean, what would prove them wrong, and where the thesis does not apply. Section 6 concludes. The appendix tabulates the test parameters.

2 Theory

2.1 Signal, noise, and the variance horizon

Consider i.i.d. trials $X_1, X_2, \ldots$ with $E[X_i] = \mu$ and $\mathrm{sd}(X_i) = \sigma$, and the partial sums $S_n = \sum_{i=1}^{n} X_i$. The accumulated signal grows linearly, $E[S_n] = \mu n$; the accumulated noise, measured by the standard deviation, grows as $\sigma\sqrt{n}$. Their ratio is the signal-to-noise ratio of the sample:

$$\mathrm{SNR}(n) = \frac{|\mu|\sqrt{n}}{\sigma}. \tag{1}$$

The edge is "visible" (comparable in magnitude to the typical fluctuation) when $\mathrm{SNR}(n) \sim 1$, i.e. at the *variance horizon*

$$n^* = \left(\frac{\sigma}{\mu}\right)^2. \tag{2}$$

n^* is a crossover *scale*, not a sharp point where the long run begins: below n^* the participant primarily experiences variance with a small drift tax; above n^* the drift dominates. To detect the edge at z standard errors requires $n \approx (z\sigma/\mu)^2$ trials. Note the quadratic punishment: halving the edge quadruples the required horizon.

2.2 The wrong-side probability: a universal asymptotic curve

For $\mu < 0$, the probability that a finite run finishes *ahead* (on the wrong side of the edge) is, by the central limit theorem,

$$P(S_n > 0) \;\approx\; \Phi\!\left(\frac{\mu n - c}{\sigma\sqrt{n}}\right) \;=\; \Phi\!\left(-\sqrt{\frac{n}{n^*}} - \frac{c}{\sigma\sqrt{n}}\right) \;\xrightarrow[n\to\infty]{}\; \Phi\!\left(-\sqrt{n/n^*}\right), \tag{3}$$

where Φ is the standard normal cdf and c is a *lattice- specific* constant, not a universal one. For the simple ± 1 lattice $c = \frac{1}{2}$ is the usual continuity correction for the strict event $S_n > 0$; for a continuous Gaussian sum the exact probability is $\Phi(-\sqrt{n/n^*})$ with no correction at all; each lattice process has its own exact threshold. The striking feature of Eq. (3) is that its large-n limit depends on the process only through the single dimensionless ratio n/n^*: a biased coin, a dice game, a roulette wheel, and a full blackjack engine all approach one curve when plotted against n/n^*. Convergence to the curve is at the Berry–Esseen rate (Petrov, 1975), and at small n lattice discreteness produces material, exactly computable deviations (verified in T1 below against exact binomial probabilities).

2.3 The zero-edge limit: the coin that never settles

For $\mu = 0$ the horizon n^* is infinite: there is no number of trials at which signal overtakes noise, because there is no signal. The typical absolute lead grows without bound,

$$E|S_n| \;\approx\; \sigma\sqrt{\frac{2n}{\pi}}, \tag{4}$$

and the fraction of time the walk spends ahead follows the arcsine law: its density is U-shaped, proportional to $[x(1-x)]^{-1/2}$, so paths spend most of their time mostly-ahead or mostly-behind and little time near even (Feller, 1968). "It evens out" is therefore true only in relative terms ($S_n/n \to 0$); in absolute terms the imbalance grows forever. A 100-flip sequence landing 60/40 against a fair coin is not an anomaly to be explained (it occurs roughly one time in twenty), but the mind's expectation that small samples resemble the population, the *law of small numbers* (Tversky and Kahneman, 1971).

2.4 Optional stopping reshapes; it does not remove

For a stopping time τ with $E[\tau] < \infty$, i.i.d. increments with finite mean, and the usual integrability conditions of the optional stopping theorem, Wald's equation gives $E[S_\tau] = \mu E[\tau]$: the expected stopped P&L equals the edge times the expected duration (Wald, 1947). Win goals and stop-losses redistribute outcomes (many small wins, a few large losses) but cannot change the mean. With $\mu < 0$, optional stopping is a reshaping machine with a fixed output mean.

2.5 Description versus inference: where the noise lives

The finite-horizon argument has a sharp boundary, and it is this: noise is not a property of data but of the *question* asked of it. A 5/10 test score, taken as a description of a closed finite universe ("I answered half the questions correctly"), is exact; there is no sampling, hence no sampling noise. The moment the same number is used to infer a latent process parameter ("am I a 50% math student?"), it becomes a sample of size $n = 10$ from the process *this student answering questions*, and the uncertainty is large, shrinking only as $1/\sqrt{n}$. Formally, with a Beta$(1, 1)$ prior on latent ability p (one of many defensible priors; the $1/\sqrt{n}$ scaling is prior-independent), the posterior after k correct of n is Beta$(1 + k, 1 + n - k)$, whose 95% credible- interval width scales as $1/\sqrt{n}$. The number didn't change; the question did.

3 Methods

Six simulation tests were run with explicit per-test seeds, 20260922 for T1 increasing by 1000 per test through 20265922 for T6, reusing a validated blackjack basic-strategy engine (2M hands: edge -0.515%, $\sigma = 1.157$ per hand, win/push/loss 43.3/8.8/47.9 (Observed, B)) where noted. Two robustness tests (T7, T8) were added with seed 20260929. Every plotted array was saved from the same run that produced the tabulated statistics. All statistics are quoted with Monte Carlo standard errors; plotted error bars are $\pm$2SE. In tables, parenthetical values are $\pm$1 Monte Carlo SE.

T1: universal collapse. Seven processes (biased coin ± 1, $p = 0.48$; dice game paying $+2$ on a six and -1 otherwise; roulette even-money; Gaussian walk $\mu = -0.05, \sigma = 1$; high-variance slot; sports bettor at -110 odds; blackjack engine), horizons at honest fractions 0.02–8 of each process's n^*, 500,000 replications per point (blackjack: up to 300,000 sessions per horizon from the earlier validated run). Recorded $P(S_n > 0)$ vs. Eq. (3), plus exact binomial probabilities for the dice process.

T2: zero-edge coin. 100,000 fair-coin paths of 1000 flips: fraction of time ahead (arcsine check); mean $|S_n|$ vs. $\sqrt{2n/\pi}$ for n up to 10^6 via direct sampling.

T3: the estimator's delusion. Losing game $\mu = -0.04$ ($p = 0.48$ coin): full distribution of the estimated edge $\hat{\mu} = S_n/n$ at eight horizons 100–20,000, 500,000 replications each; fraction with $\hat{\mu} > 0$.

T4: volume asymmetry. "House" trajectories of $10^4/10^5/10^6/10^7$ bets vs. 20,000 "players" $\times \{50, 200, 1000\}$ bets, same $p = 0.48$ coin; final *and* pathwise (maximum) statistics.

T5: description vs. inference. Beta$(1, 1)$ posterior 95% intervals for latent ability after $n \in \{10, \ldots, 10^6\}$ observations at exactly the true rate.

T6: optional stopping, generalized. $p = 0.48$ random walk, 100,000 paths; Rule A: stop at $+20/-50$; Rule B: stop at $+10$ or a 5,000-step cap; Rule C: stop at $+50/-200$; all three rules share the 5,000-step backstop cap (binding for 0.03% of A paths, 55% of B, 42% of C). Check $E[S_\tau] = \mu E[\tau]$ and tabulate full stopped-P&L distributions.

T7: fat-tailed increments. Student-t increments with $\nu \in \{3, 5\}$, normalized to marginal $\sigma = 1$, $\mu = -0.04$ (so $n^* = 625$ by construction, identical to the Gaussian baseline). Wrong-side rates at fractions 0.02–8 of n^*, 60,000 replications per point, plus an iid Gaussian baseline at the same replication count.

T8: autocorrelated trials. AR(1) increments $X_i - \mu = \rho(X_{i-1} - \mu) + \varepsilon_i$ with $\rho \in \{+0.5, -0.5\}$, marginal $\sigma = 1$. The long-run variance is $\sigma^2_{\mathrm{LR}} = \sigma^2(1+\rho)/(1-\rho)$, so the effective horizon is $n^*_{\mathrm{LR}} = n^*_{\mathrm{iid}}(1+\rho)/(1-\rho)$ (1875 for $\rho = +0.5$, 208 for $\rho = -0.5$). Wrong-side rates at fractions 0.02–8 of n^*_{LR}, 30,000 replications per point.

Cohort analog. 20,000 players × 200 bets under each of the T7/T8 increment processes (same μ, σ footing), recording the fraction finishing ahead: the T4 success rate under violated assumptions.

Evidence grading. Every reported number carries an evidence tag. *Observed* denotes a quantity measured from a primary source or a validated archive; *Modelled* a quantity computed from the simulation pipeline; *Assumed* a quantity fixed by judgment. Source quality follows in parentheses: A, official primary source; B, reputable secondary source or validated engine; C, provisional. The 2M-hand blackjack validation figures are Observed, B; every test-battery statistic is Modelled, B. Uncertainty is stated in the same sentence as each headline number as ±1 Monte Carlo standard error (in tables, parenthetical).

Retrospective analysis. The simulation battery was completed before this paper was written, and no analysis plan was pre-registered.

4 Results

4.1 T1: seven games, one (asymptotic) curve

Figure 1 shows the collapse. Plotted against n/n^*, the biased coin ($n^* = 624$), dice game ($n^* = 5$), roulette even-money ($n^* = 360$), Gaussian walk ($n^* = 400$), slot ($n^* = 819$), −110 sports bettor ($n^* = 441$), and blackjack ($n^* = 50{,}463$) all follow $\Phi(-\sqrt{n/n^*})$. Maximum |sim − theory| is 0.13 at the smallest horizons (dice, $n = 2$), falling under 0.02 for $n/n^* \geq 2$ for every process; for blackjack it is 0.014 (all Modelled, B; Monte Carlo SEs below 0.001). The horizon spans four orders of magnitude across games, yet the curve is one. At $n = n^*/4$ the wrong-side probability is still ≈31%; even at $n = n^*$ it is ≈16%.

The curve is asymptotic, and the deviations are informative rather than embarrassing. For the dice process the simulation matches the *exact binomial* wrong-side probability to within 0.16 percentage points at every horizon (the worst case across all six simulated processes), while the smooth theory curve deviates from the exact values by up to 13.4 points at $n \leq 5$ (Figure 1, crosses). Discreteness, not noise, explains the small-n gap: the universal curve is the large-n limit, and each lattice knows its own exact form.

4.2 T2: the coin that never settles

Figure 2 confirms the degenerate case. The fraction of time ahead is U-shaped: 19.3% of fair-coin paths spend over 90% of 1000 flips ahead, 21.4% spend under 10% ahead (the small asymmetry is ties, counted as not-ahead under the strict $>$); 28.4% spend over 80% ahead and 30.4% under 20%: one side leading "most of the time" is the norm, not the exception (Modelled, B; SEs under 0.2pp). Mean absolute lead tracks $\sqrt{2n/\pi}$ out to 10^6 flips (796.6 vs. 797.9). In absolute terms the imbalance grows without bound.

4.3 T3: a losing game looks winning

For the $\mu = -4\%$ game, the estimated edge $\hat{\mu}$ at $n = 200$ exceeds zero 26.2% of the time (Modelled, B; SE 0.06pp) and exceeds +2% in 17.9% of runs (Figure 3). The delusion decays on the universal curve: 17.3% at $n = 500$, 9.7% at $n = 1000$ (2.6% still show $> +2\%$), 3.5% at $n = 2000$, 0.24% at $n = 5000$, 0.002% at $n = 10000$ (12 of 500,000 paths, matching the

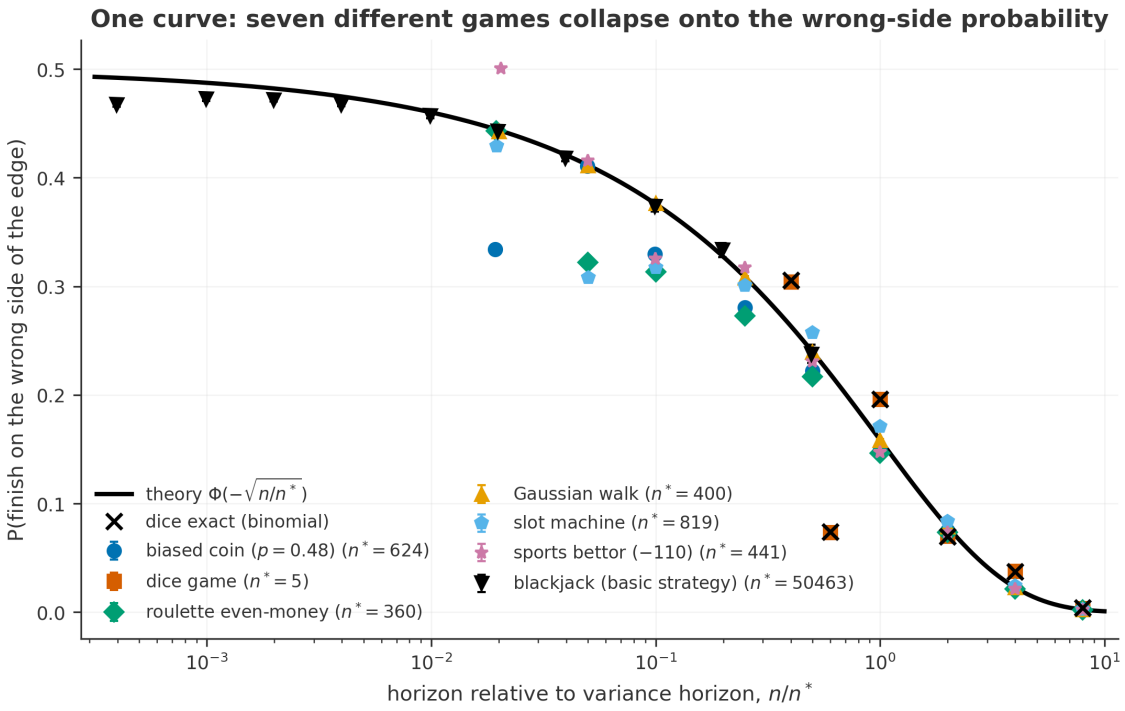


Figure 1: Wrong-side probability vs. horizon relative to n^*. Seven unrelated processes collapse onto the asymptotic curve $\Phi(-\sqrt{n/n^*})$. Crosses: exact binomial probabilities for the dice process: simulation matches them to 0.16pp, confirming the small-n deviations are lattice discreteness, not simulation error.

theoretical $\Phi(-4.008) \approx 3 \times 10^{-5}$), and zero of 500,000 at $n = 20000$. A "winning player" over a few hundred trials of a losing game is not beating the odds: they are sampling the right tail of the estimator's distribution.

4.4 T4: same game, two statistical universes

Figure 4: the house's realized edge converges as $1/\sqrt{N}$: −3.30% (1.00 pp) at 10^4 bets, −3.93% (0.32 pp) at 10^5, −3.99% (0.10 pp) at 10^6, −3.97% (0.03 pp) at 10^7 (parenthetical: ±1SE) against a truth of −4% (all Modelled, B). The 20,000 players, each with 200 bets, finish with mean −8.1 units and standard deviation 14.2 units; 26.0% finish ahead. Identical process, identical edge, and two non-overlapping statistical realities, separated only by volume.

The players' table sharpens the point (Table 1). *Final* and *pathwise* statistics must be distinguished: at 200 bets, 11.5% of players *finish* at +10 or better, but 29.8% *touch* +10 at some point; the longer the session, the wider the gap grows (at 1000 bets: 6.2% finish ≥ +10, while 42.8% touch it). Selection on the path, not the endpoint, manufactures winners.

Table 1: 20,000 players, $p = 0.48$ coin. Final vs. pathwise outcomes.

bets	mean (u)	finish ahead	finish ≥+10	ever ≥+10	ever ≥+25
50	−2.0	33.4%	5.9%	10.5%	0.0%
200	−8.1	26.0%	11.3%	29.9%	2.6%
1000	−40.1	10.0%	6.3%	42.8%	11.4%

Notes: Percentages are cohort fractions; Monte Carlo SEs are under 0.35pp for every cell. "Ever" columns are pathwise maxima over the session.

Source: Kilambi Research simulation archive (`results.json`); all values Modelled, B.

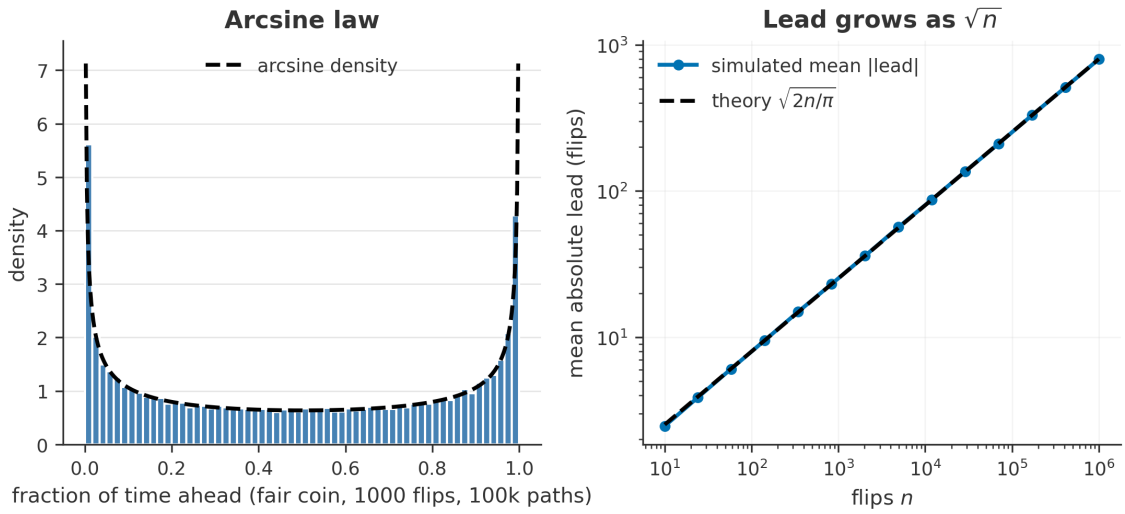


Figure 2: Zero-edge coin. Left: arcsine law for the fraction of time ahead (1000 flips, 100k paths). Right: mean absolute lead grows as $\sqrt{n}$ without bound, verified to 10^6 flips.

The advertised winner is an order statistic. The best final result of the 20,000-player cohort at 200 bets was +50 units, the kind of outcome that gets advertised. It is not evidence of skill: the predicted cohort maximum (exact-binomial median of the maximum of 20,000 draws) is $\approx +48$ units, and the observed +50 agrees. At 1000 bets the prediction is $\approx +86$ vs. +106 observed, a +20-unit overshoot. The maximum is itself noisy: Gumbel fluctuations of order ± 7 units at that horizon (Leadbetter et al., 1983) make +106 roughly a 1-in-25 realization ($\approx 4\%$), still no evidence of skill. With enough players, someone *must* post a spectacular score; the cohort maximum is a deterministic consequence of cohort size, and marketing selects it.

4.5 T5: the math-test point, quantified

Table 2 gives the 95% credible interval for latent ability p (truth 0.5) after n observations at exactly the true rate. As *description*, k/n is exact at every n; as *inference*, the interval at $n = 10$ is $[0.23, 0.77]$ (the test says almost nothing about ability), narrowing as $1/\sqrt{n}$ to $[0.499, 0.501]$ at $n = 10^6$.

Table 2: Posterior 95% interval for latent ability ($p_{\text{true}} = 0.5$).

n	observed	95% CI	width
10	5/10	[0.23, 0.77]	0.53
100	50/100	[0.40, 0.60]	0.19
1000	500/1000	[0.47, 0.53]	0.062
10000	5000/10000	[0.490, 0.510]	0.020
10^6	500000/10^6	[0.4990, 0.5010]	0.0020

Notes: Beta(1,1) prior; intervals computed exactly from the Beta posterior. Widths scale as $1/\sqrt{n}$.

Source: Kilambi Research simulation archive (`results.json`); computed quantities, Modelled, B.

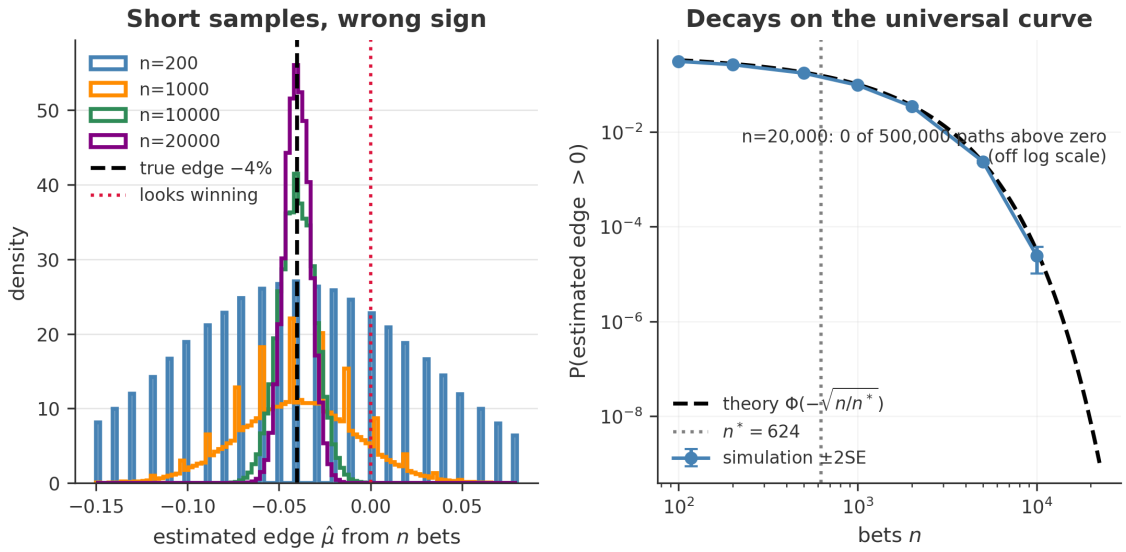


Figure 3: Distribution of the estimated edge $\hat{\mu}$ for a $\mu = -4\%$ game at four horizons. Short samples routinely estimate the wrong sign; the fraction doing so decays along $\Phi(-\sqrt{n/n^*})$.

4.6 T6: stopping rules, generalized

Wald's equation holds across rules (Table 3). Rule A hits the +20 goal in 20.0% of sessions yet averages −36.0 units (0.2), matching $\mu E[\tau] = -36.1$. Rule B (stop at +10 or a 5,000-step cap) wins 45.0% of sessions and still averages −114.6 units (0.8) versus −114.5 predicted. Rule C (+50/−200) wins only 1.9% and averages −169.8 versus −169.9 (0.3). The backstop cap binds for 55% of B paths and 42% of C paths (0.03% of A), which is why those rules run thousands of steps; Wald's equation holds regardless, since the cap makes τ bounded (all Modelled, B). A near-coin-flip win rate coexists with a certain, large mean loss: stopping rules reshape, never remove.

Table 3: Optional stopping: $E[S_\tau]$ vs. $\mu E[\tau]$ ($\mu = -0.04$), 100,000 paths.

Rule	$E[S_\tau]$	$E[\tau]$	$\mu E[\tau]$	P(hit goal)
A: +20 / −50	−36.0 (0.2)	902	−36.1	0.200
B: +10 / cap 5000	−114.6 (0.8)	2864	−114.5	0.450
C: +50 / −200	−169.8 (0.3)	4248	−169.9	0.019

Notes: Parenthetical values are ±1 Monte Carlo SE of the stopped mean. $E[\tau]$ and $\mu E[\tau]$ are exact to the quoted digits.

Source: Kilambi Research simulation archive (`results.json`); all values Modelled, B.

4.7 T7–T8: the curve survives fat tails and autocorrelation

The universality claim of T1 was proved for i.i.d. trials. Two violations were tested: heavy-tailed increments (T7) and serially correlated trials (T8). In both cases the test is the same as T1: does $P(S_n > 0)$ follow $\Phi(-\sqrt{n/n^*})$ when plotted against the horizon?

Fat tails (T7). Student-t increments with $\nu = 3$ (infinite third moment; kurtosis undefined) deviate from the curve by at most 1.37 percentage points across all horizons (Modelled, B; worst case at $n/n^* = 0.02$, Monte Carlo SE 0.20pp), and by at most 0.04pp for $n/n^* \geq 2$. For $\nu = 5$ the maximum deviation is 0.38pp. The iid Gaussian baseline at the same replication count deviates by at most 0.15pp. Fat tails slow the convergence slightly at very short horizons but do

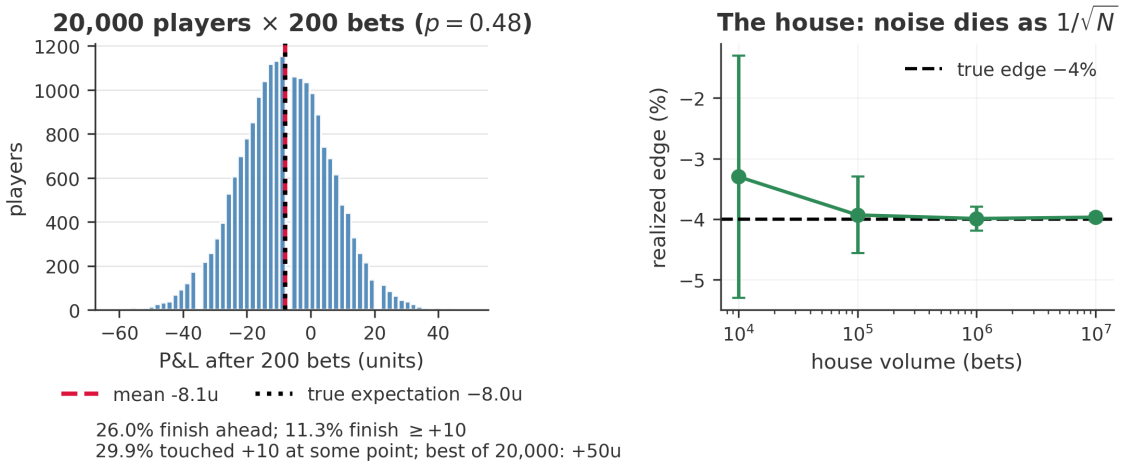


Figure 4: Volume asymmetry. 20,000 players × 200 bets of a $p = 0.48$ coin: a wide spray with 26% winners. The house, with up to 10^7 bets, measures its edge to ±0.03pp.

not move the curve: the claim that heavy tails only lengthen the effective horizon is, if anything, conservative here, since with σ held fixed the curve is essentially unchanged.

Correlated trials (T8). For AR(1) increments the naive *iid* horizon is the wrong ruler: with $\rho = +0.5$ the long-run variance is 3× the marginal variance, so variance dominance sets in three times later ($n^*_{\mathrm{LR}} = 1875$ vs. 625). Plotted against the long-run horizon, the collapse holds: the collapse holds to within Monte Carlo noise at every horizon (maximum deviation 0.27pp against a per-point SE of 0.29pp; Modelled, B). With $\rho = -0.5$ the horizon shortens by a factor of three ($n^*_{\mathrm{LR}} = 208$) and the collapse holds on the shortened horizon: maximum deviation 1.14pp, confined to the shortest horizon ($n = 10$, an AR startup transient); from $n/n^* \geq 0.1$ the deviations are at most 0.45pp, and 0.24pp for $n/n^* \geq 2$ (Modelled, B). The $1/\sqrt{n}$ logic is untouched; correlation redefines the ruler, exactly as the long-run-variance formula says it must.

Cohort success rates under violated assumptions. Table 4 reports the T4-style cohort (20,000 players × 200 bets) under each increment process on the same $\mu = -0.04, \sigma = 1$ footing, alongside the collapse deviations.

Table 4: Robustness: wrong-side collapse deviations and 200-bet cohort success rates under fat-tailed and autocorrelated increments.

Increment process	max |sim − theory| (all n/n^*)	max |sim − theory| ($n/n^* \geq 2$)	P(finish ahead, 200 bets)
iid Gaussian (baseline)	0.15pp	0.12pp	28.5 (0.3)
Student-t, $\nu = 5$	0.38pp	0.07pp	28.4 (0.3)
Student-t, $\nu = 3$	1.37pp	0.04pp	27.8 (0.3)
AR(1), $\rho = +0.5$	0.27pp	0.25pp	36.8 (0.3)
AR(1), $\rho = -0.5$	1.14pp	0.24pp	17.0 (0.3)

Notes: Theory is $\Phi(-\sqrt{n/n^*})$ with n^* from long-run variance (1875 for $\rho = +0.5$, 208 for $\rho = -0.5$, 625 otherwise). Deviations in percentage points. Cohort: 20,000 players × 200 bets; Monte Carlo SEs under 0.35pp.

Source: Kilambi Research simulation archive (`results_robust.json`, `tests_robust.py`); all values Modelled, B.

The honest reading: neither violation breaks the thesis, but positive autocorrelation weakens the participant's position further (the horizon lengthens, so at a fixed 200 bets more players finish ahead of a losing edge: 36.8 (0.3) vs. 28.5 (0.3) baseline), while negative autocorrelation

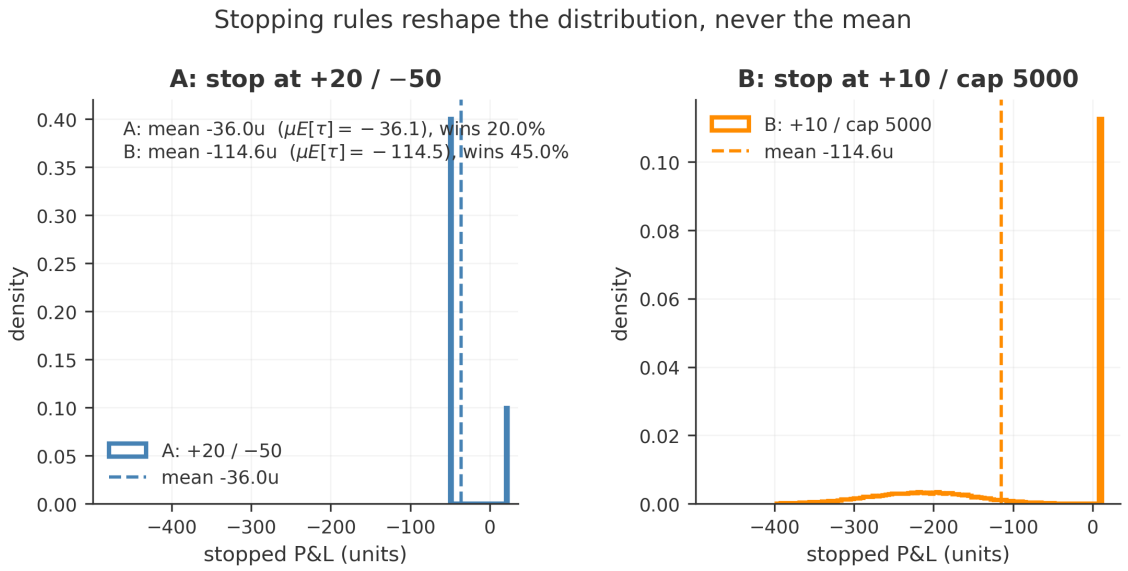


Figure 5: Stopped-P&L distributions under rules A and B (100k paths; histograms and quoted statistics from the same run). Mass piles at the goal, the tail absorbs the rest, the mean stays at $\mu E[\tau]$.

shortens the horizon and lets the edge show sooner. The universality claim, stated on the long-run horizon, survives both.

5 Discussion

The odds belong to whoever has the volume. T4 is the paper's applied core. The casino does not need any individual to lose; it needs the *sum* over millions of hands, where its horizon is $n \gg n^*$ and only the edge survives. The player, confined to $n \ll n^*$, gets the variance. The same asymmetry structures insurance (the insurer pools; the insured is $n = 1$), markets (the market index is the ensemble; the day trader is a short path), and sportsbooks (volume vs. the weekend bettor). In each case the published number (the premium, the expected return, the -110) is the process parameter of the aggregator, not a description of any constituent's trajectory. This is the finite-horizon face of the ergodicity-economics observation (Peters, 2019): ensemble averages and time averages describe different objects, and individuals live on time trajectories.

Why the intuition misfires. The mind expects small samples to mirror the population (Tversky and Kahneman, 1971): 100 flips "should" look 50/50, a 200-hand session "should" cost about 1 unit. T2 and T3 show the opposite is typical: skewed coins, winning losers, sessions that live far from even. The published odds are not wrong; they are addressed to the process, and no individual trajectory *is* the process.

Falsifiability: what would prove this wrong. Three sharp falsifiers. First, the universal curve: a new finite-variance process whose wrong-side rates deviate from $\Phi(-\sqrt{n/n^*})$ by more than a few percentage points at $n/n^* \geq 2$, with n^* computed from the process's own long-run variance, would break the universality claim (T1, T7, and T8 currently bound the deviations at 1.4pp (Modelled, B)). Second, the variance-horizon scale: repeated demonstrations of an edge being reliably detectable (say, correct sign in 95% of independent samples) at $n \ll n^*/10$ would show n^* is not the governing scale. Third, optional stopping: a stopping rule with $E[S_\tau] \neq \mu E[\tau]$ under the stated regularity conditions would overturn the reshaping result; the

paper's claim is exactly that no such rule exists. The description–inference boundary is falsified by a closed-universe description that carries sampling noise, or by inference from n samples that beats the $1/\sqrt{n}$ rate without additional structure.

Scope and limits. The thesis bites exactly where finite samples are used for inference: judging a betting system on 500 bets, a trading strategy on 40 trades, a diet on two weeks, a candidate on one interview round. It does not bite where the finite set *is* the universe and nothing is inferred: final scores, census counts, the 5/10 as a record. T7 and T8 locate the boundary of the universality claim precisely: fat tails (to $\nu = 3$) and AR(1) autocorrelation (to $|\rho| = 0.5$) move n^* through the long-run variance but leave the $1/\sqrt{n}$ logic and the asymptotic curve intact. Regime changes and model misspecification remain outside the tested scope: a process whose μ or σ shifts mid-sample has no single n^*, and the tests here do not cover it. All results are for stationary trials; the universality claim is for the large-n limit, with exact small-n forms available per process.

6 Conclusion

The long run begins, as a crossover scale, at $n^* = (\sigma/\mu)^2$ trials, and almost nobody gets there. Below it, outcomes are variance with a drift tax, wrong-side finishes approach one universal asymptotic curve (verified under fat tails and autocorrelation in T7–T8), zero-edge processes never settle, and stopping rules only rearrange the losses. Whether finite data carries noise at all depends on the question: description is exact, inference is noisy. The actionable moral is correspondingly sharp: never infer a process from a sample small against its horizon, notice who in any transaction holds the volume, and remember that the most advertised winner is usually just the largest order statistic of the largest cohort.

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A Test parameters

Table 5: Simulation test parameters.

Process	μ	σ	n^*	horizons (n/n^*)
Biased coin (± 1, $p = 0.48$)	-0.040	0.999	624	0.02–8
Dice (+2 on six, -1 else)	-0.500	1.118	5	0.4–8 (honest)
Roulette even-money	-0.053	0.999	360	0.02–8
Gaussian walk	-0.050	1.000	400	0.02–8
Slot machine	-0.100	2.862	819	0.02–8
Sports bettor (-110)	-5.0	105.0	441	0.02–8
Blackjack, basic strategy	-0.00515	1.157	50,463	4×10^{-4}–0.5

T2: fair coin, 100,000 paths $\times$ 1000 flips; lead growth to $n = 10^6$.

T3: $p = 0.48$ coin, $n \in \{100, \ldots, 20000\}$, 500,000 replications.

T4: house 10^4–10^7 bets; 20,000 players $\times$ $\{50, 200, 1000\}$ bets, $p = 0.48$.

T5: Beta$(1, 1)$ posterior, $n \in \{10, \ldots, 10^6\}$.

T6: $p = 0.48$ walk, 100,000 paths, rules +20/$-$50, +10/cap-5000, +50/$-$200.

T7: Student-t increments $\nu \in \{3, 5\}$, $\mu = -0.04$, $\sigma = 1$, $n^* = 625$, 60,000 replications per horizon; iid Gaussian baseline at same count.

T8: AR(1) increments $\rho \in \{+0.5, -0.5\}$, marginal $\sigma = 1$; $n^*_{\mathrm{LR}} = 1875$ ($\rho = +0.5$), 208 ($\rho = -0.5$); 30,000 replications per horizon.

Source: Kilambi Research simulation archive (`results.json`, `results_deep.json`, `results_robust.json`); all values Modelled, B.